

第3章 环烃单元测验（临床类）

1

关于苯结构的描述正确的是：

（多选 分）

A.

苯分子为平面正六边形

（正确答案）

B.

所有碳原子均为 sp^2 杂化，各杂化轨道完全相同。

（正确答案）

C.

苯分子中的闭合的共轭大 π 键非常稳定。

（正确答案）

D.

苯是两种重要共振式的共振杂化体；其两种共振式相互转变的速率极快。

（错误答案）

2

芳香烃的亲电取代反应的中间体是：

（单选 分）

A.

碳正离子

（正确答案）

B.

碳负离子

（错误答案）

C.

碳自由基

（错误答案）

D.

环闭合共轭体系

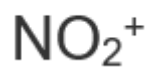
（错误答案）

3

苯硝化反应的亲电试剂是：

(单选 分)

A.



(正确答案)

B.



(错误答案)

C.



(错误答案)

D.



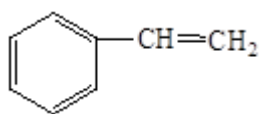
(错误答案)

4

分子中所有的C都属于 sp^2 的是:

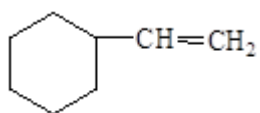
(单选 分)

A.



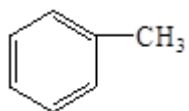
(正确答案)

B.



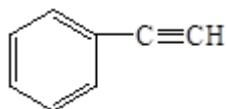
(错误答案)

C.



(错误答案)

D.



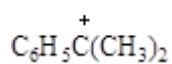
(错误答案)

5

下面碳正离子结构最稳定的是：

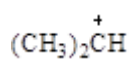
(单选 分)

A.



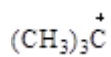
(正确答案)

B.



(错误答案)

C.



(错误答案)

D.



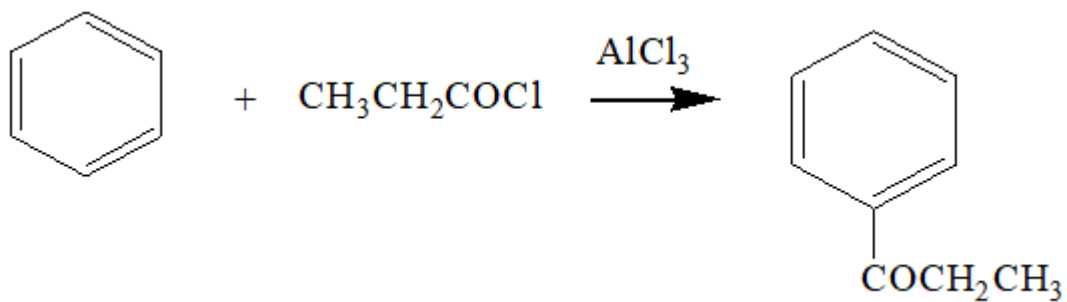
(错误答案)

6

下面反应式正确的是：

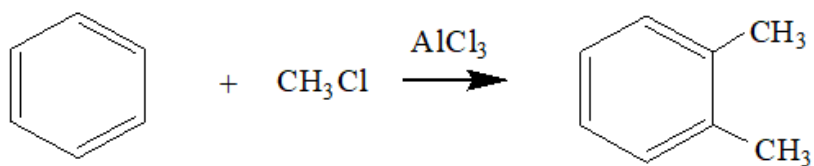
(单选 分)

A.



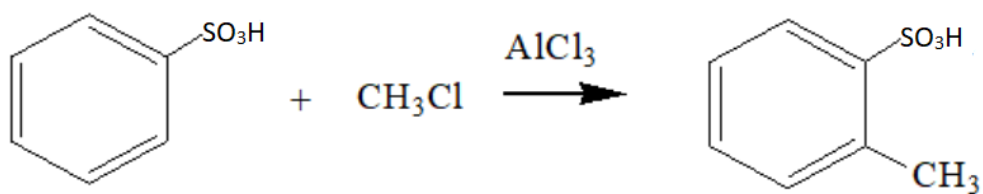
(正确答案)

B.



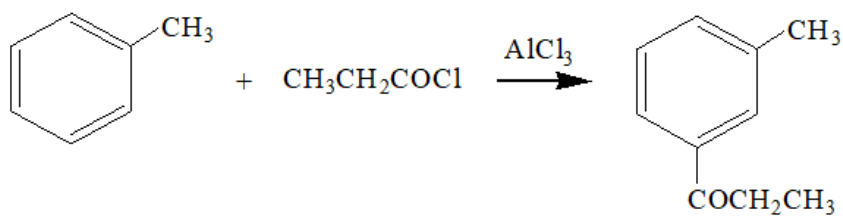
(错误答案)

C.



(错误答案)

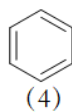
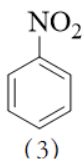
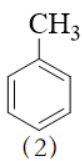
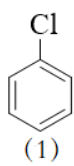
D.



(错误答案)

7

下列化合物发生芳香亲电取代反应，反应活性顺序由高到低的顺序是：



(单选 分)

A.

(2) > (4) > (1) > (3)

(正确答案)

B.

(1) > (4) > (3) > (2)

(错误答案)

C.

(4) > (2) > (3) > (1)

(错误答案)

D.

(3) > (4) > (2) > (1)

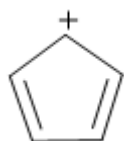
(错误答案)

8

下面有机化合物无芳香性的是：

(单选 分)

A.



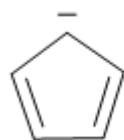
(正确答案)

B.



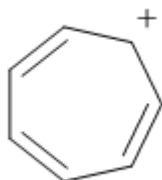
(错误答案)

C.



(错误答案)

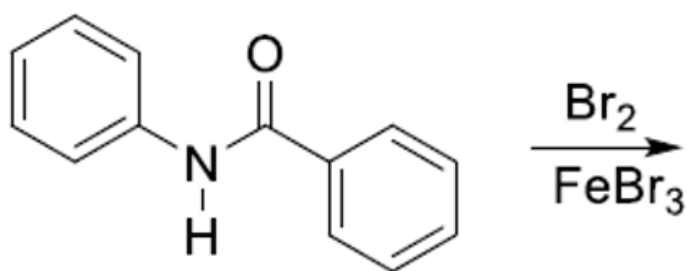
D.



(错误答案)

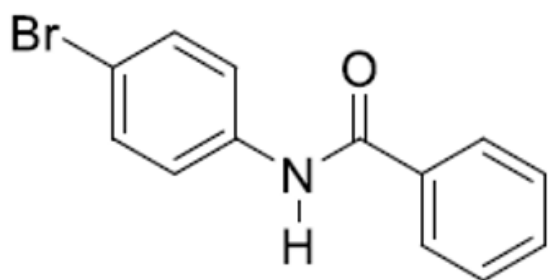
9

下列反应生成的主要产物是：



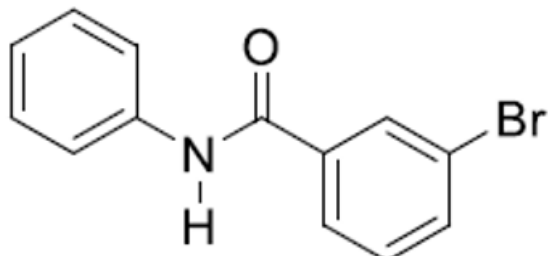
(单选 分)

A.



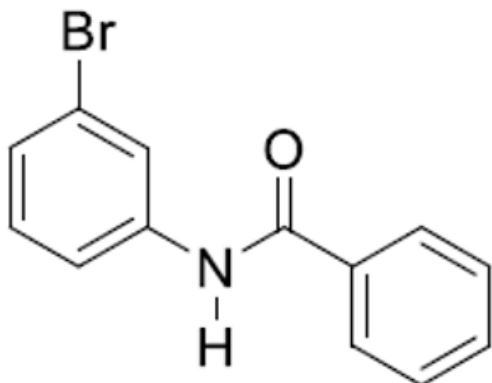
(正确答案)

B.



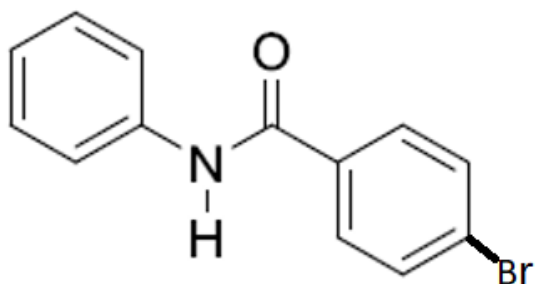
(错误答案)

C.



(错误答案)

D.



(错误答案)

10

分子环张力最大的是：

(单选 分)

A.

环丙烷

(正确答案)

B.

环丁烷

(错误答案)

C.

环戊烷

(错误答案)

D.

环己烷

(错误答案)

11

区别环丙烷和丙烷可用下列哪种试剂？

(单选 分)

A.

Br₂

(正确答案)

B.

KMnO₄

(错误答案)

C.

浓 H₂SO₄

(错误答案)

D.

HBr

(错误答案)

12

区别环丙烷和丙烯可用下列哪种试剂？

(单选 分)

A.

KMnO₄

(正确答案)

B.

Br₂

(错误答案)

C.

HCl

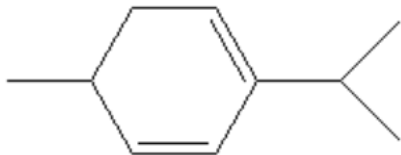
(错误答案)

D.

[Ag(NH₃)₂]⁺

(错误答案)

13



的 IUPAC (1980 年版) 命名是：

(单选 分)

A.

5-甲基-2-异丙基-1,3-环己二烯

(正确答案)

B.

3-甲基-6-异丙基-1,5-环己二烯

(错误答案)

C.

4-甲基-1-异丙基-1,5-环己二烯

(错误答案)

D.

6-甲基-3-异丙基-1,3-环己二烯

(错误答案)

14

甲基环丙烷与 HBr 反应的产物是：

(单选 分)

A.

2-溴丁烷

(正确答案)

B.

1-溴丁烷

(错误答案)

C.

3-溴丁烷

(错误答案)

D.

2-丁烯

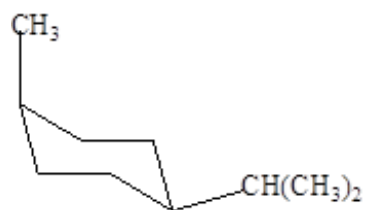
(错误答案)

15

顺-1-甲基-4-叔丁基环己烷最稳定的构象是：

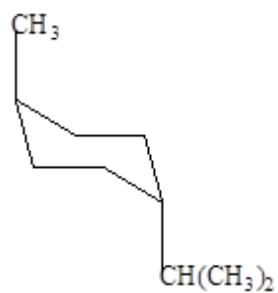
(单选 分)

A.



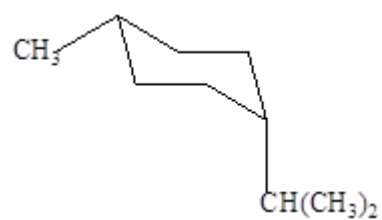
(正确答案)

B.



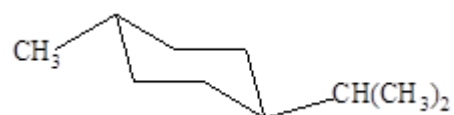
(错误答案)

C.



(错误答案)

D.



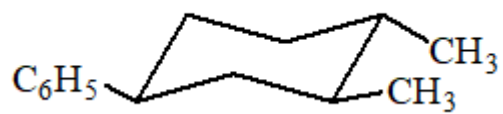
(错误答案)

16

某化合物的几种构象如下，其优势构象为：

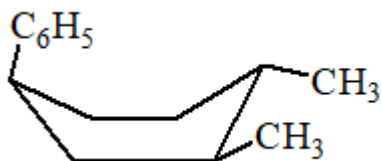
(单选 分)

A.



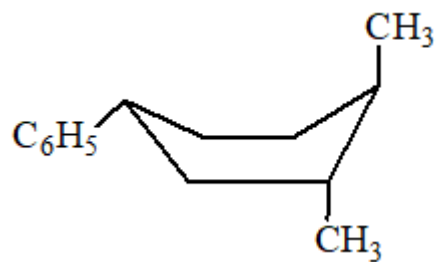
(正确答案)

B.



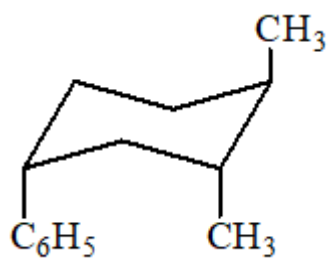
(错误答案)

C.



(错误答案)

D.



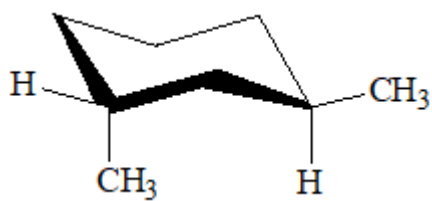
(错误答案)

17

反-1, 3-二甲基环己烷的优势构象为:

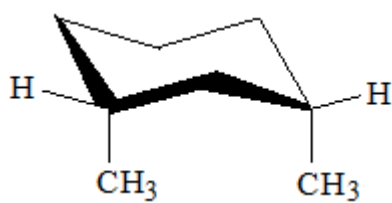
(单选 分)

A.



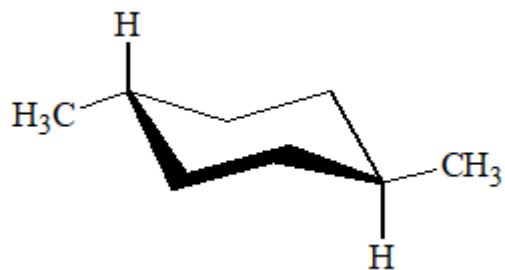
(正确答案)

B.



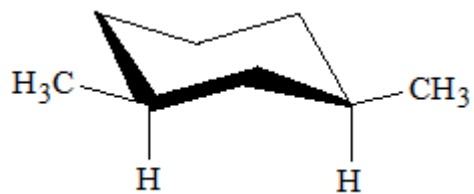
(错误答案)

C.



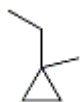
(错误答案)

D.



(错误答案)

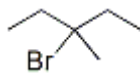
18



与 HBr 反应的主要产物是:

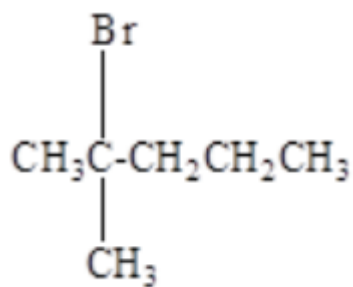
(单选 分)

A.



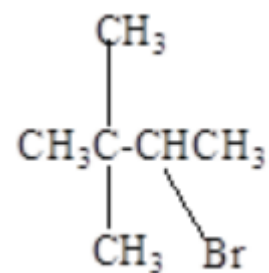
(正确答案)

B.



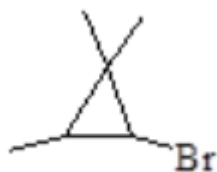
(错误答案)

C.



(错误答案)

D.



(错误答案)

19

顺-十氢萘与反-十氢萘的异构属于哪种异构形式？

(单选 分)

A.

顺反异构

(正确答案)

B.

位置异构

(错误答案)

C.

构象异构

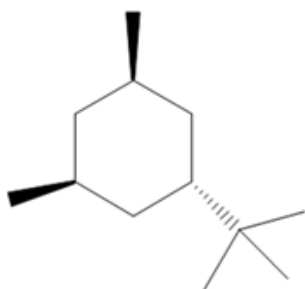
(错误答案)

D.

对映异构

(错误答案)

20



最稳定的构象中：

(单选 分)

A.

两个甲基都在 a 键

(正确答案)

B.

两个甲基都在 e 键

(错误答案)

C.

叔丁基在 a 键

(错误答案)

D.

一个甲基在 a 键，一个甲基在 e 键

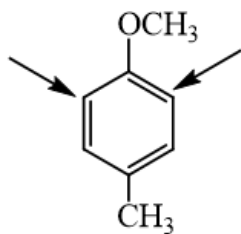
(错误答案)

21

下列化合物进行磺化反应时，第三个基团磺酸基主要进入的位置正确的是：

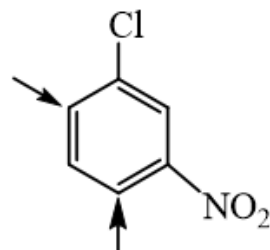
(多选 分)

A.



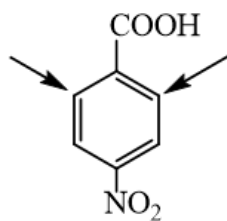
(正确答案)

B.



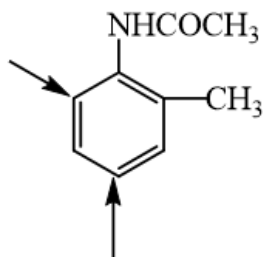
(正确答案)

C.



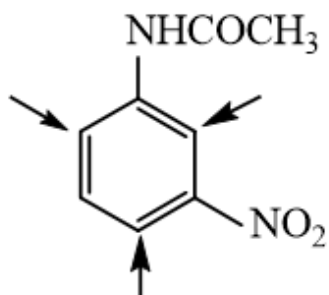
(正确答案)

D.



(正确答案)

E.



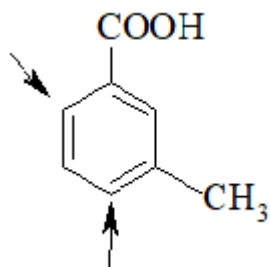
(错误答案)

22

下列化合物进行硝化反应时，第三个基团硝基主要进入的位置正确的是：

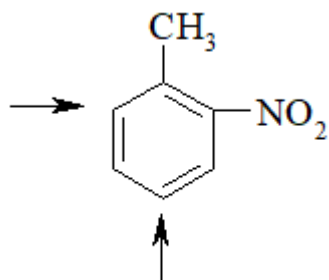
(多选 分)

A.



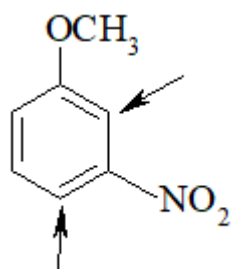
(正确答案)

B.



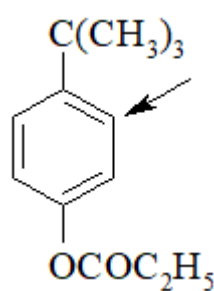
(正确答案)

C.



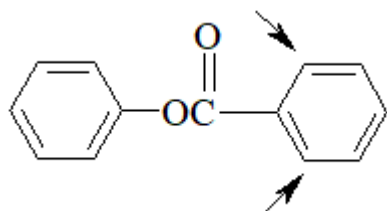
(错误答案)

D.



(错误答案)

E.



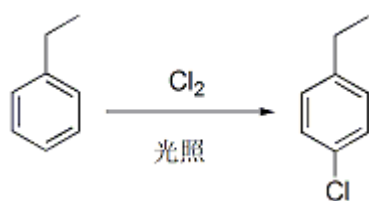
(错误答案)

23

下面反应式**错误**的是：

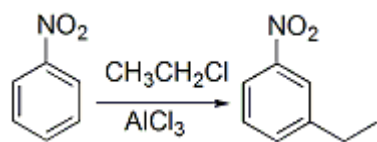
(多选 分)

A.



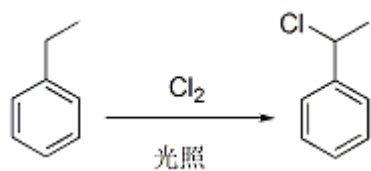
(正确答案)

B.



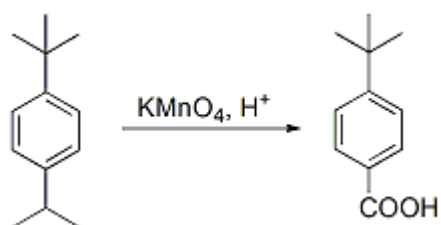
(正确答案)

C.



(错误答案)

D.



(错误答案)